



Data Dimension Reduction

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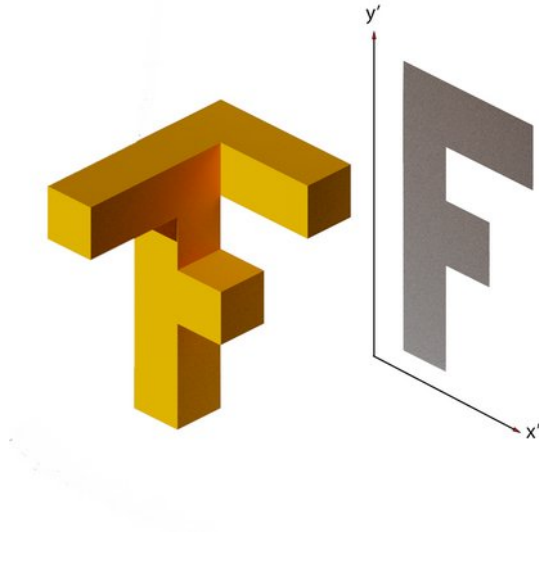


Agenda for today

- This Chapter
 - Data Dimension Reduction overview (today)
 - Concepts behind Data Reduction and PCA
 - Importance of rescaling data
 - Hands on data reduction application in Python (next lecture)
 - Intro to non-linear dimensionality reduction (Just a taste, next lecture)

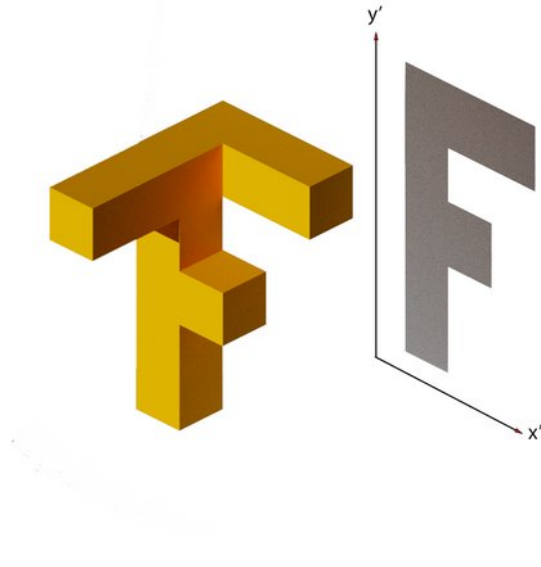
Intuition – Project your data to a lower Dim Space

- The difficulty is in reducing the data, while preserving the signal/patterns underlying it



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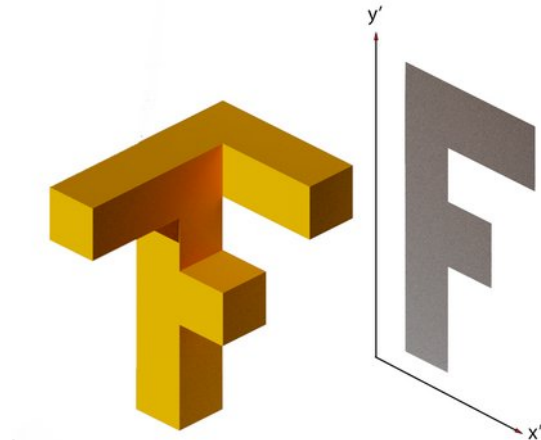


3D shape

x	y	z
0.74	0.06	0.08
0.89	0.89	0.92
0.66	0.94	0.60
0.93	0.14	0.89
0.40	0.83	0.73
0.28	0.54	0.59
0.33	0.04	0.35
0.99	0.54	0.78
0.87	0.73	0.85
...	...	...

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0.33	0.04	0.35
0.99	0.54	0.78
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...	...	...

PCA >>>

2D shape projected

x'	y'
0.73	0.62
0.83	0.00
0.17	0.79
0.41	0.05
0.80	0.97
0.52	0.07
0.93	0.88
0.39	0.32
0.97	0.11
...	...

Data Dimension Reduction

- Data reduction is an important component of data analysis
 - Objective 1: **Structure understanding**:
 - Are some variables similar to each other, and if so, can we somehow group them together?
 - Objective 2: **Improving efficiency**:
 - Reducing data size while preserving the important information, for more efficient subsequent analysis.
- Differences: data summarization vs. data reduction
 - Data **summarization** “compresses the **rows**”
 - Data **reduction** preserves the “rows”, but tries to compress the “**columns**”
- Intuitive Examples:
 - Categorical variables can be reduced by combining similar categories
 - “enjoys snowboarding”, “likes skiing” & “likes hockey” can be combined into “winter sports fan”
 - Numerical variables can be transformed into a smaller set of weighted averages of the original data that contain most of the original information in less variables
 - “GMAT quant score”, “Math SAT” & “Math GPA” can be combined into a single “Quant ability”

Issues with high-dimensional data

- High-dimensional data suffers from the “curse of dimensionality”
 - **Sparsity increases:** Distance between points, which is critical for models like clustering and outlier detection, become less meaningful.
 - **Overfitting:** The **number of samples needed** to produce meaningful insights **increases** exponentially. Having many features but not enough data could lead to worse performance
- Also...
 - Too many variables unnecessarily complicate **interpretation** of analysis
 - How do you visualize 100-dimensional data?
 - Too many variables make **multicollinearity** more likely, which may **severely impact the effectiveness** with some of the methods we will study in this course.

Example of curse of dimensionality – exponential data

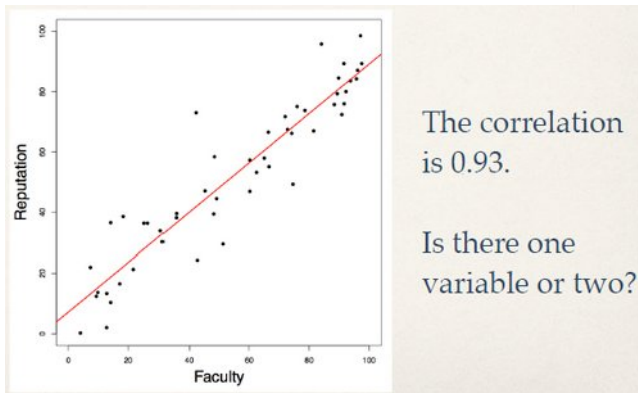
- Let's pretend we're trying to build a hiring model and have a dataset that only contains simple binary (yes/no) categories:
 - Cat 1: college degree? (yes/no)
 - Cat 2: rookie or seasoned? (rookie/seasoned)
 - Cat 3: top 10% of high-school class? (yes/no)
 - ...
- For statistical soundness, assume we need at least 10 observations for each possible outcome
 - Dataset with only cat 1: 10 “yes” and 10 “no” = 20 samples needed
 - Dataset with cat 1 and cat 2: 4 unique combinations: yes & rookie, yes & seasoned, no & rookie, no & seasoned. If each needs 10 samples, we need 40 total samples.
 - Dataset with cat 1, 2 and 3: 8 unique combinations... 80 total sample points
 - ...
 - Dataset with “n” categories, we need $10 * 2^n$ total samples!
 - For $n=250$, we need: 1,000,000...,000,000 (75 zeros!, ~ # of atoms in universe)

A popular technique for data reduction is PCA

- **Principal Component Analysis (PCA)** is a descriptive (exploratory) approach that involves projections (like the intro slide)
- The objective is to reduce a set of numerical variables (columns) by removing any overlap of information between variables
- Final product: A smaller number of numerical variables that (hopefully) contain most of the information
- How does PCA do this?
 - Rely on correlations to create new variables that are linear combinations of the original variables (i.e., they are weighted averages of the original variables)
 - These linear combinations are uncorrelated (no information overlap), and only a few of them contain most of the original information
 - The new variables are called principal components (or “factors” or “features” or “super variables”)

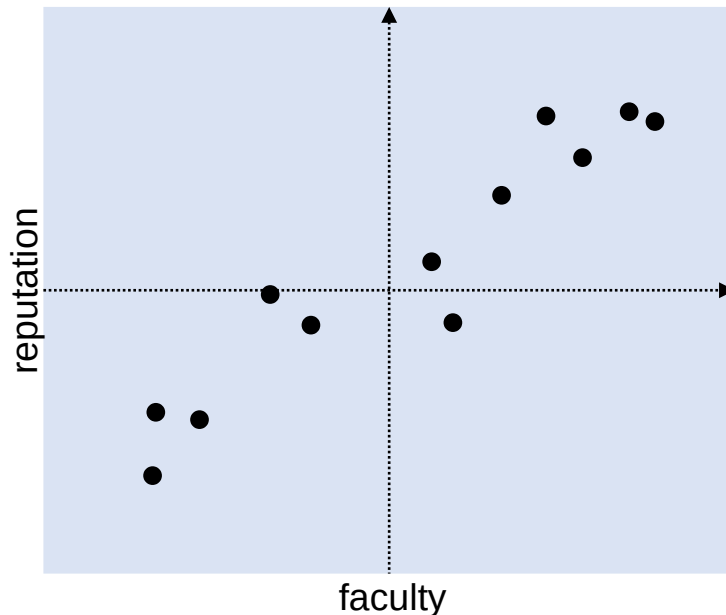
Example #1: 2D MBA Ratings

- Fifty MBA students rated the importance of “quality of faculty” and “reputation of school” in their choice of business school. Assume that is the prediction goal (y).
- Are these two variables measuring something similar?
 - Heuristically, yes, if quality of faculty is correlated with reputation of school
- Can the number of variables be reduced from two to one?
 - Heuristically, yes (because they are very highly correlated).



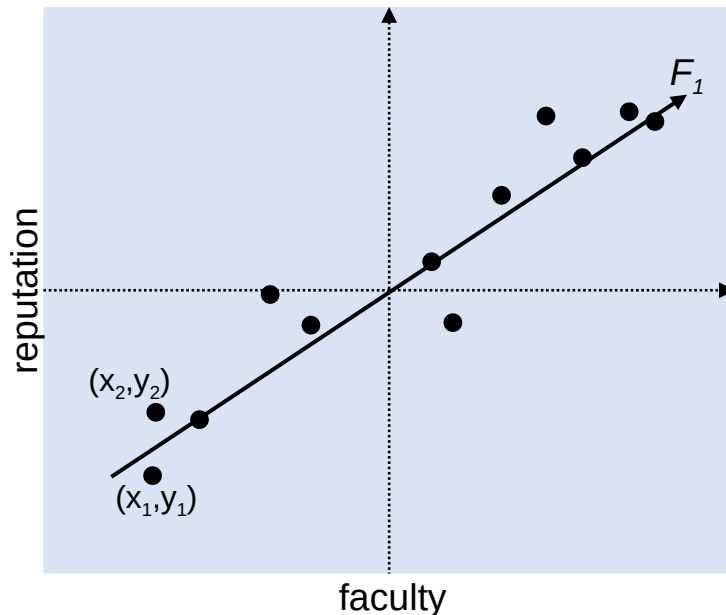
Example #1: 2D MBA Ratings

- Principal components
 - factor F1 is the first principal component
 - factor F2 is the second
 - factor F2 is orthogonal to F1
- PC's are projections...
 - What does F1 represent?
 - A combination...
- How does projection achieve dimension reduction?
 - A point (x_1, y_1) in original coordinate system becomes z_1 (single number) when projected onto F_1
 - (x_2, y_2) becomes z_2 , etc...
- There are as many "F's" as columns



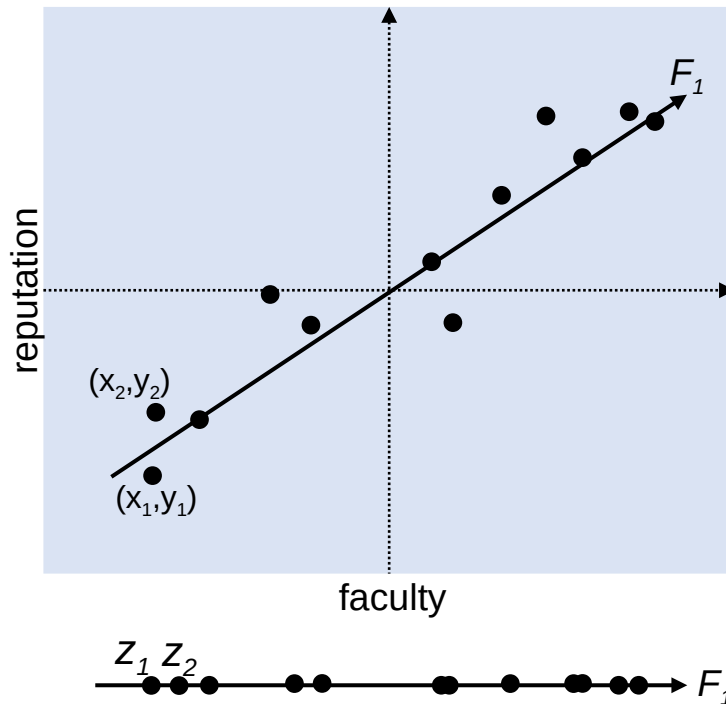
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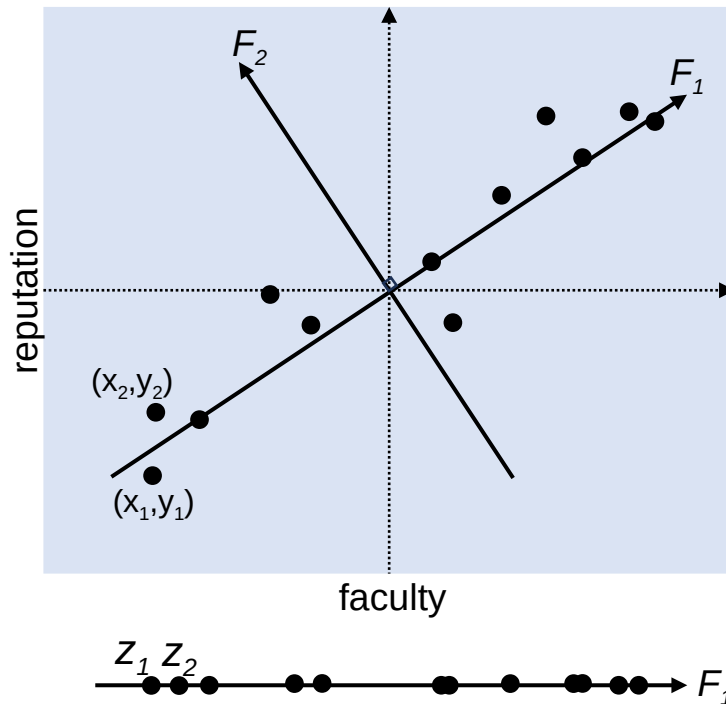
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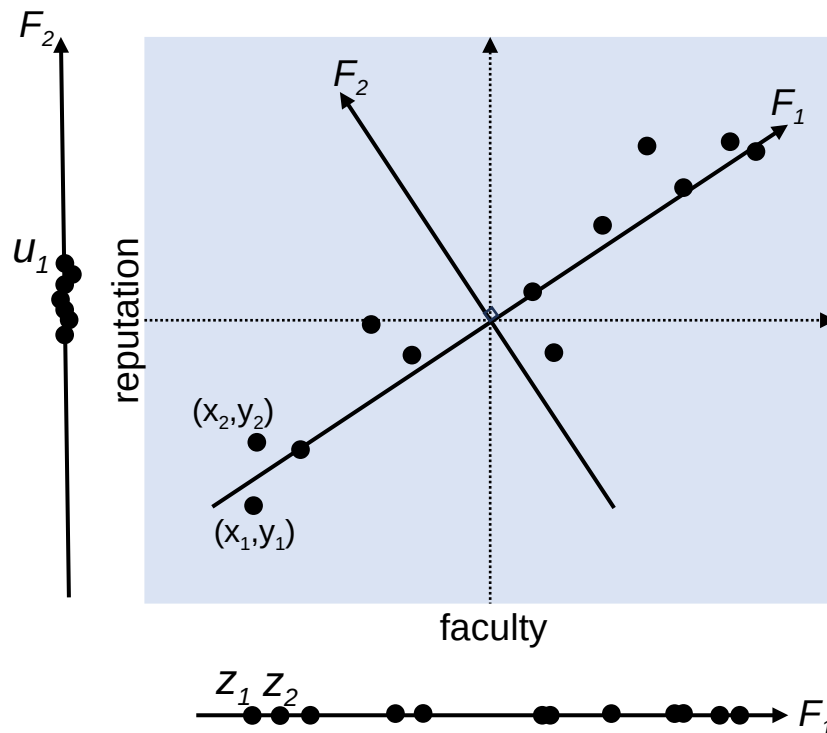
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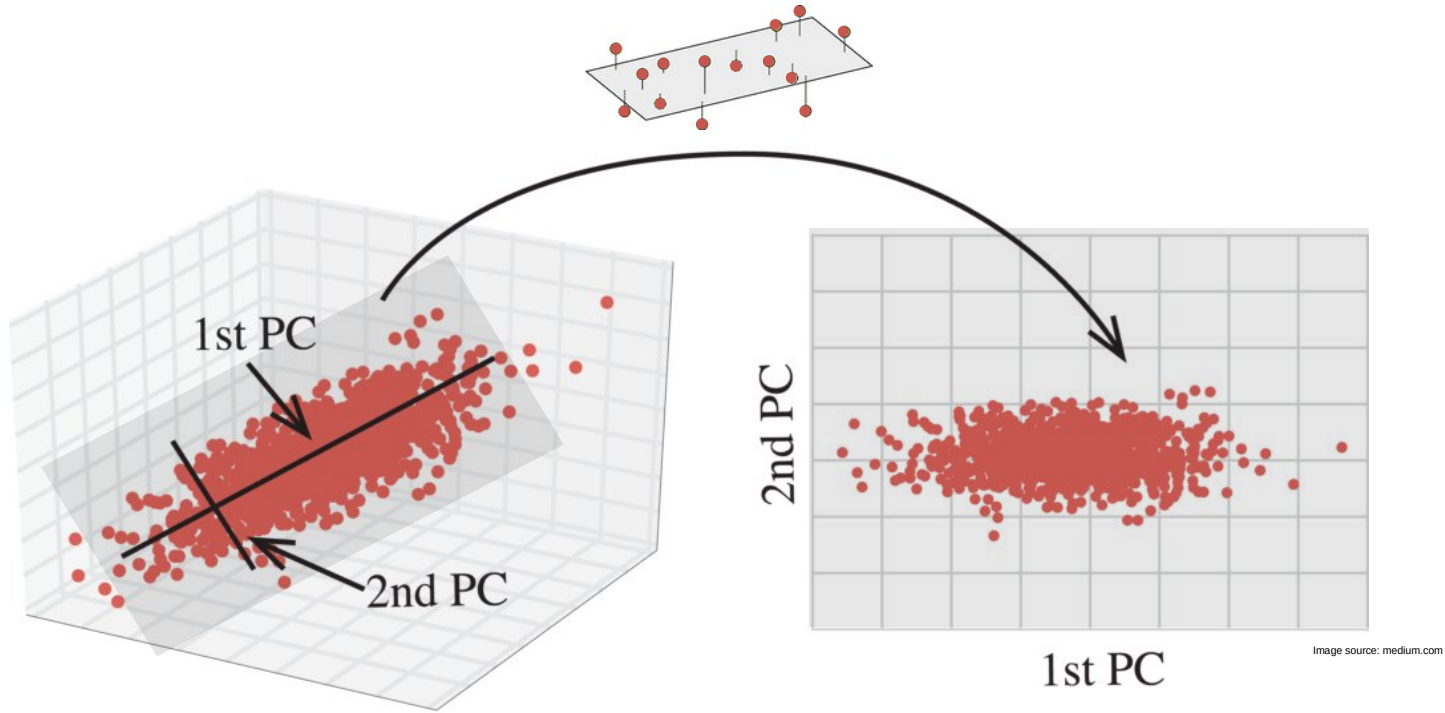


Properties of PCs

- *PC1* lies along the direction along which the data varies the most.
- All PCs are perpendicular to each other (i.e., uncorrelated).
- PCs are ordered in terms of their ability to explain the data (i.e., from the highest to the lowest variance).
- Let's play with [this](#)

3D, 4D, ..., 1000D etc.

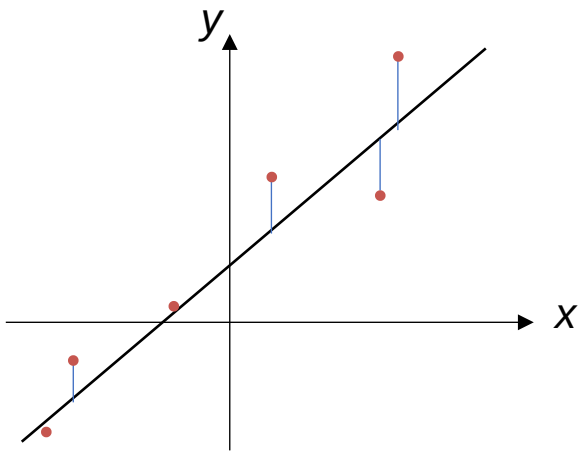
- Same idea holds...



Careful: PCA is not the same as Linear Regression!

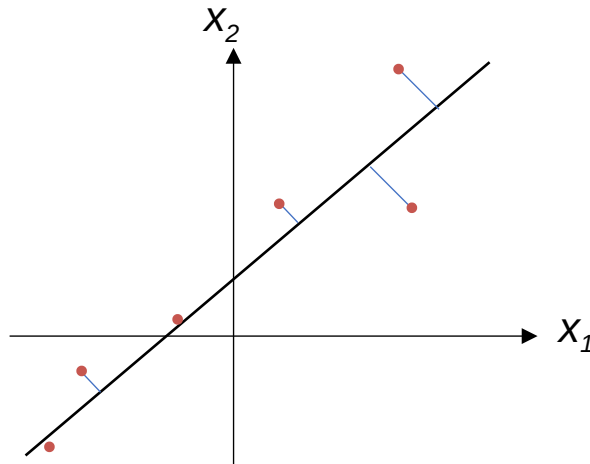
- Linear regression

- y variable is “special”
 - $y = f(x)$



- PCA

- x_1 and x_2 are equal features
 - Projection angle is different: PC aims to maximize variance of proj. points



- The two methods have different objectives and can lead to very different outcomes

PCA algorithm – Data Preprocessing

Before running PCA, we need to “process” the data to turn all relevant data into metric data, and rescale:

- PCA uses correlations -> Performed on metric variables
 - A small number of “dummy variables” can be included in a set of metric variables that are analyzed
- Scaling issues
 - Some variables may have greater scale than others
 - E.g., compare “house size” vs. “# of bedrooms”
 - Their variance will be a dominant component of the total variance
 - Rescale variables to remove the scale effect, otherwise measurement units affect results
 - Normalization: Rescale the variables into a range of [0,1]
 - Standardization: Rescale the variables to have a mean of 0 and a standard deviation of 1

PCA Output

- Component weights (in component matrix) reflect the correlation between a particular variable and the component
 - Correlations should ideally (hopefully?) be high
- There are as many components as there are dimensions in the data (columns)
- If a component is correlated with a number of features, that can help with interpreting that component at a higher level.

Principal component loadings

California housing block groups (n = 20,640) · all variables standardized (z-scores)

	Component							
	1	2	3	4	5	6	7	8
Median income	0.086	-0.058	0.922	0.370	-0.020	-0.018	0.037	-0.004
Median house age	-0.429	0.025	-0.407	0.806	0.014	0.026	0.009	-0.001
Total rooms	0.956	0.100	0.102	0.104	0.120	0.162	-0.119	0.015
Total bedrooms	0.970	0.083	-0.121	0.056	0.144	-0.068	0.051	-0.083
Population	0.933	0.034	-0.121	0.076	-0.327	0.034	0.006	-0.015
Households	0.972	0.086	-0.113	0.087	0.058	-0.112	0.061	0.083
Latitude	-0.140	0.970	0.017	-0.088	0.017	0.132	0.113	0.005
Longitude	0.144	-0.969	-0.062	-0.063	0.037	0.136	0.109	0.007

Extraction: principal component analysis on the correlation matrix; 8 components extracted.

Shaded cells mark loadings of |0.40| or greater — the variables that define each component.

PCA Output

- The eigenvalues represent the magnitude of the components
 - If data is standardized, each variable has variance ~ 1
 - An eigenvalue of 3.901 means this component has the explicatory power of almost 4 variables! (call it a supervariable!)
- The % of variance in the data explained by component i is computed by taking the ratio of its eigenvalue, to the sum of eigenvalues.

$$\% \text{ explained} = \frac{\lambda_i}{\lambda_1 + \dots + \lambda_m}$$

Variance explained by each component

Initial eigenvalues · California housing block groups (n = 20,640)

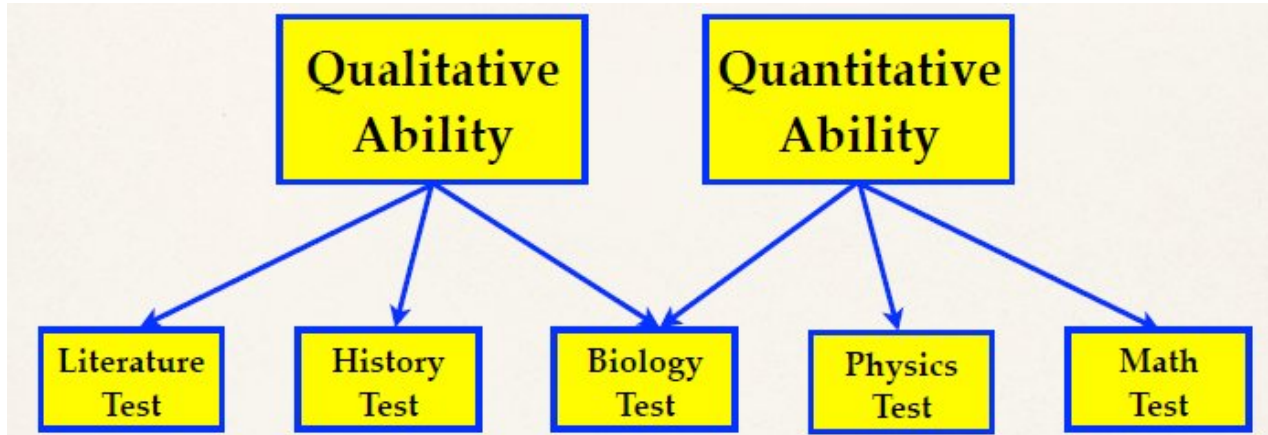
Component	Eigenvalue	% of variance	Cumulative %
1	3.901	48.767	48.767
2	1.910	23.881	72.648
3	1.073	13.409	86.057
----- eigenvalue = 1 · Kaiser cut-off			
4	0.825	10.311	96.368
5	0.148	1.847	98.215
6	0.082	1.020	99.235
7	0.047	0.586	99.821
8	0.014	0.179	100.000

Extraction: principal component analysis on the correlation matrix.

The first 3 components carry 86.1% of the variance in the 8 original variables.

Example of Components

- One might imagine that humans have two basic abilities, qualitative and quantitative.
- While one cannot measure these abilities directly, one can take measurements on quantities which should “derive” from them (some will derive from one, some from the other, and some from both).
- If our theory is true, these measurements will be correlated.



How do we choose the Number of Components?



- Common stopping criteria:
 - **A Priori Criterion:** A pre-determined number of components based on the business objective and constraints or prior experience
 - **Latent Root Criterion:** Components with eigenvalues greater than 1
 - **Percentage of Variance Criterion:** Enough components to meet a specified percentage of variance explained, usually 70% or higher
 - **Scree Test Criterion:** Components before inflection point
- Consider several alternative solutions (one more and one less component than the initial solution) to ensure the best structure is identified

Eigenvalue Plot for Scree Test Criterion

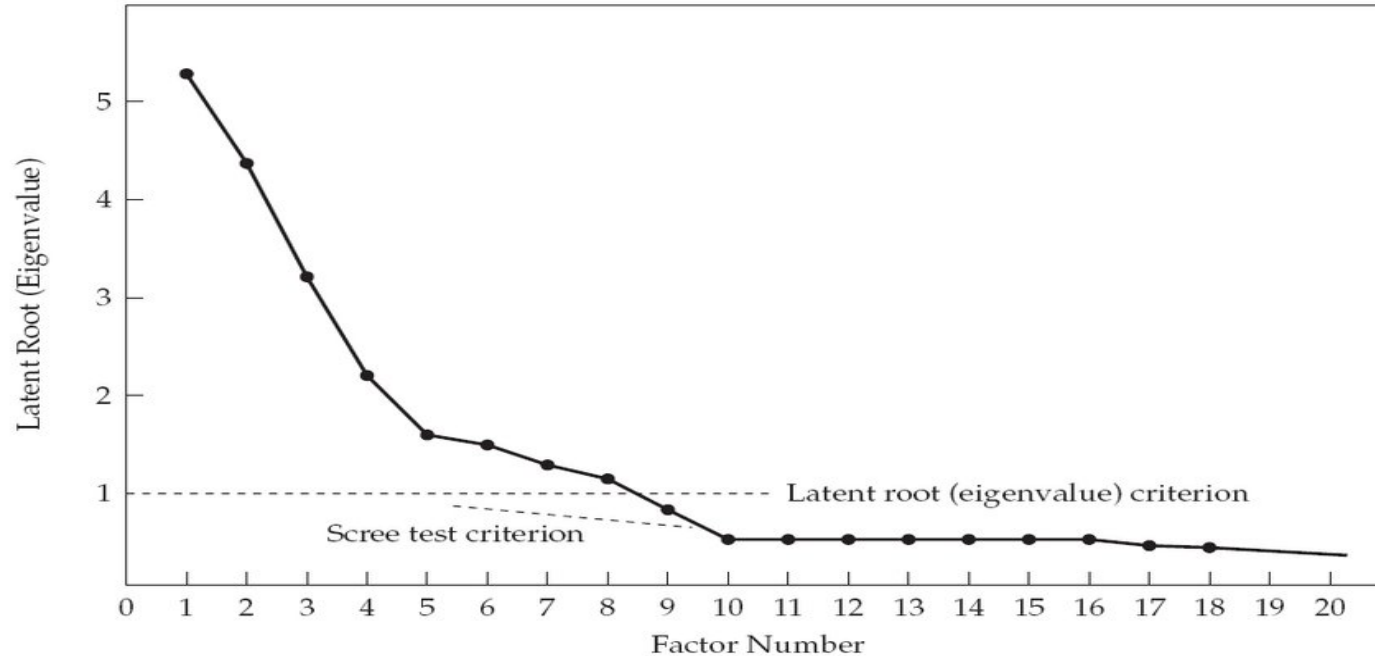


FIGURE 3-6 Eigenvalue Plot for Scree Test Criterion

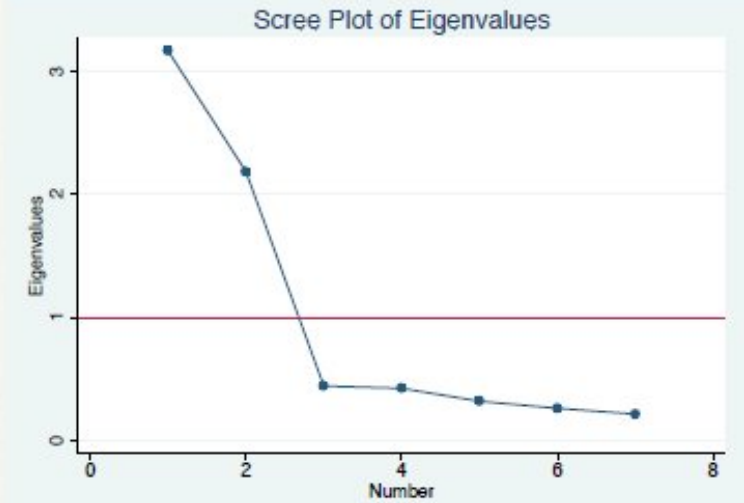
Example #2: Shopping

- Shopping Questionnaire: A local mall collected survey data from 212 respondents. Each respondent was asked to rate the following statements using a ten-point scale (0 = disagree; 10 = agree):
 - Q1: Shopping is fun.
 - Q2: Shopping is bad for your budget.
 - Q3: I combine shopping with eating out.
 - Q4: I try to get the best buys while shopping.
 - Q5: I don't care about shopping.
 - Q6: You can save a lot of money by comparing prices.
 - Q7: I can't spend enough time shopping.
- Are there really 7 useful dimensions here?

How Many Components/Factors are there?

- This dataset has two eigenvalues above one suggesting that there are two factors in the data
- Dimension can be reduced from 7D to 2D!

Eigenvalue #	Eigenvalue	Prop. of Var.	Cum. Prop. of Var.
1	3.16386	0.4520	0.4520
2	2.18376	0.3120	0.7639
3	0.44082	0.0630	0.8269
4	0.42186	0.0603	0.8872
5	0.31714	0.0453	0.9325
6	0.25967	0.0371	0.9696
7	0.21289	0.0304	1.0000



Interpreting the Components/Factors

- Possible interpretation of the 2 main factors: **Enjoyment** and **Bargain Hunting**

Q1: Shopping is fun.

Q2: Shopping is bad for your budget.

Q3: I combine shopping with eating out.

Q4: I try to get the best buys while shopping.

Q5: I don't care about shopping.

Q6: You can save a lot of money by comparing prices.

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```
. kfactor q*, rotation(varimax)  
(obs=212)
```

Factor analysis/correlation
Method: principal-component factors

Number of obs = 212
Retained factors = 2
Number of params = 13

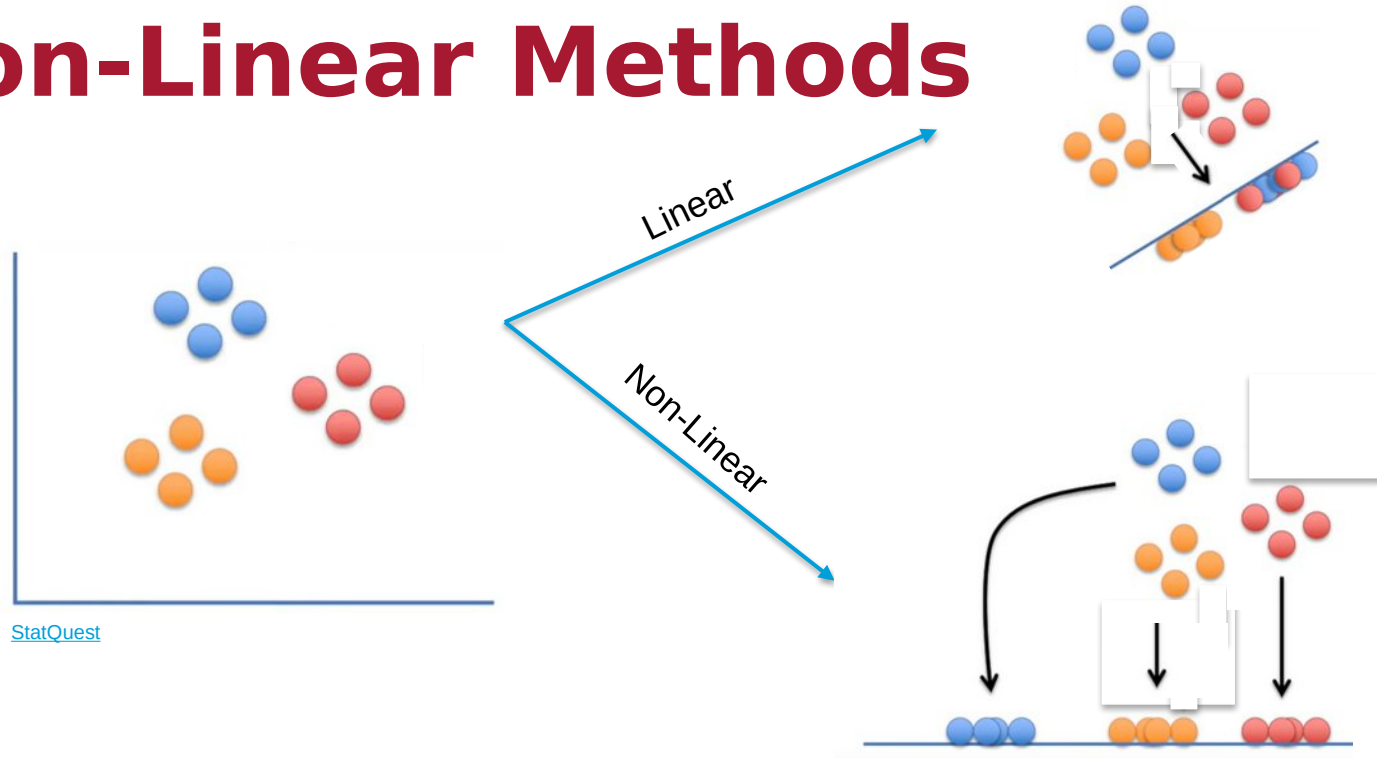
Rotated factor loadings (pattern matrix) and unique variances sorted

Variable	Factor1	Factor2	Uniqueness
q1	0.8951	0.0718	0.1936
q3	0.8865	0.0624	0.2102
q5	-0.8738	-0.0075	0.2365
q7	0.8303	0.0639	0.3066
q2	0.0001	0.8959	0.1974
q6	0.0343	0.8881	0.2101
q4	0.1183	0.8295	0.2980

Where PCA sits among the alternatives

Method	Best For	Handles Numeric?	Handles Categorical?	Intuition	Key Strengths	Key Weaknesses
PCA (covered in BA305)	Numeric data with linear relationships	✓ Yes	✗ only few	Finds new rotated axes that capture the most variance	Simple, removes redundancy, widely used	Only captures linear relationships, sensitive to scaling
MCA (Multiple Correspondence Analysis) — not covered	Categorical data (one-hot encoded)	✗ No	✓ Yes	Finds latent categories that explain similarities between responses	Great for survey data	Doesn't handle numerical features
FAMD (Factor Analysis for Mixed Data) — not covered	Mixed numerical & categorical data	✓ Yes	✓ Yes	Balances PCA (numeric) & MCA (categorical) in one model	Best for mixed datasets, prevents one type from dominating	More complex, requires proper scaling
t-SNE (t-Distributed Stochastic Neighbor Embedding)	Clustering & visualizing high-dimensional data	✓ Yes	✓ Yes	Groups similar points together based on distance relationships	Great for visualizing clusters	Computationally expensive, not good for feature reduction
UMAP (Uniform Manifold Approximation and Projection)	Fast visualization & clustering	✓ Yes	✓ Yes	Like t-SNE, but preserves more global structure	Faster than t-SNE, works well for high dimensions	Hard to interpret feature relationships

Non-Linear Methods



Visual Example

- UMAP and tSNE are two methods for non-linear dimensionality reduction.

- [General Demo](#)
- [UMAP](#)
- [tSNE](#)